Uniform Locality Reduction for the Descent-Paired Staircase: The Entire T-System Family Reduces to Single Base Cases

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Abstract

Let $\Pi_q^{(k)} = B_1 B_2 \cdots B_k$ where $B_j = (1+T_j)(1+T_{j-1})\cdots(1+T_1)$ in the Iwahori–Hecke algebra $H_q(S_n)$, and let $M^{(k)}[D_n, D_n]$ be the matrix of right multiplication by $\Pi_q^{(k)}$ restricted to the descent-paired set $D_n \subset S_n$. We prove, uniformly in all parameters, that for every even $K \geq 4$, every $k \in \{1, \ldots, K-1\}$, and every $n \geq K$:

$$\det M^{(k)}[D_n, D_n] \;=\; \big(\det M^{(k)}[D_K, D_K]\big)^{|D_n|/|D_K|}, \qquad |D_n| = n!/2^{\lfloor n/2 \rfloor}.$$

This generalises the $K = 4$ locality reduction of the first-pair theorem (Apr 18) and the $K = 6$ locality reduction of the second-pair theorem (Apr 19) into a single uniform statement covering all even K.

As an immediate corollary, the conjectured T-system identity

$$\det M^{(2j+1)}[D_n, D_n] \cdot \det M^{(2j-1)}[D_n, D_n] \;=\; q^{|D_n|}\big(\det M^{(2j)}[D_n, D_n]\big)^2$$

at any (j, n) with $n \geq 2j+2$ reduces, unconditionally, to the single base case at $n = 2j+2$. In particular, the $j = 3$ identity holds for all $n \geq 8$ provided only that it holds at $n = 8$, which is supported computationally by 9/9 mod-prime tests (companion note `2026-04-20-tsystem-j3-evidence.tex`).

1 Setup

Definition 1 (Hecke algebra conventions)**.** $H_q(S_n)$ is the Iwahori–Hecke algebra over $\mathbb{Z}[q]$ with generators $T_1, \ldots, T_{n-1}$, quadratic relation $T_i^2 = (q-1)T_i + q$, and the usual braid relations. Its $\mathbb{Z}[q]$-basis is $\{T_w : w \in S_n\}$. Right multiplication by T_i acts as

$$T_w \cdot T_i \;=\; \begin{cases} T_{ws_i}, & w(i) < w(i+1), \\ q\,T_{ws_i} + (q-1)\,T_w, & w(i) > w(i+1), \end{cases}$$

where ws_i swaps positions i and $i+1$ in the one-line notation of w.

Definition 2 (Staircase elements)**.** $B_j := (1+T_j)(1+T_{j-1})\cdots(1+T_1)$ and $\Pi_q^{(k)} := B_1 B_2 \cdots B_k$ for $1 \leq k \leq n-1$.

Definition 3 (Descent-paired set)**.** $D_n := \{w \in S_n : w(2i-1) > w(2i) \text{ for all } 1 \leq i \leq \lfloor n/2 \rfloor\}$, of cardinality

$$|D_n| \;=\; \frac{n!}{2^{\lfloor n/2 \rfloor}}.$$

Definition 4 (Staircase matrix)**.** For $u, v \in D_n$ and $k \in \{1, \dots, n-1\}$,

$$M^{(k)}[u, v] \;:=\; [T_u]\big(T_v \cdot \Pi_q^{(k)}\big) \in \mathbb{Z}[q],$$

the coefficient of T_u in $T_v \cdot \Pi_q^{(k)}$. This defines a $|D_n| \times |D_n|$ matrix over $\mathbb{Z}[q]$.

For a permutation $w \in S_n$ and an integer $m \in \{0, \dots, n\}$, write $\tau_m(w) := (w(m+2), \dots, w(n))$ for the length-$(n-m-1)$ suffix of the one-line notation. (When $m = n-1$, $\tau_m(w)$ is empty.)

2 Main Theorem and Corollary

Theorem 5 (Uniform locality reduction)**.** *Let $K \geq 4$ be even. For every $k \in \{1, \dots, K-1\}$ and every $n \geq K$,*

$$\det M^{(k)}[D_n, D_n] \;=\; \big(\det M^{(k)}[D_K, D_K]\big)^{|D_n|/|D_K|}$$

as polynomials in $\mathbb{Z}[q]$, where $|D_n|/|D_K|$ is a non-negative integer.

Corollary 6 (Conditional T-system at all n)**.** *Fix $j \geq 1$, and set $K := 2j+2$. Suppose the base-case identity*

$$\det M^{(2j+1)}[D_K, D_K] \cdot \det M^{(2j-1)}[D_K, D_K] \;=\; q^{|D_K|}\big(\det M^{(2j)}[D_K, D_K]\big)^2 \tag{1}$$

holds in $\mathbb{Z}[q]$. Then for every $n \geq K$,

$$\det M^{(2j+1)}[D_n, D_n] \cdot \det M^{(2j-1)}[D_n, D_n] \;=\; q^{|D_n|}\big(\det M^{(2j)}[D_n, D_n]\big)^2.$$

Proof of Corollary 6 given Theorem 5. Apply Theorem 5 with $K = 2j+2$ to each $k \in \{2j-1, 2j, 2j+1\}$ (all lie in $\{1, \dots, K-1\}$). Let $m := |D_n|/|D_K|$. Then

$$\begin{aligned}
\det M^{(2j+1)}[D_n] \cdot \det M^{(2j-1)}[D_n] &= \big(\det M^{(2j+1)}[D_K]\big)^m \cdot \big(\det M^{(2j-1)}[D_K]\big)^m \\
&= \big(\det M^{(2j+1)}[D_K] \cdot \det M^{(2j-1)}[D_K]\big)^m \\
&\overset{(1)}{=} \big(q^{|D_K|}(\det M^{(2j)}[D_K])^2\big)^m \\
&= q^{m\,|D_K|}\big(\det M^{(2j)}[D_K]\big)^{2m} \\
&= q^{|D_n|}\,\big((\det M^{(2j)}[D_K])^m\big)^2 \\
&= q^{|D_n|}\,\big(\det M^{(2j)}[D_n]\big)^2,
\end{aligned}$$

where the last line uses Theorem 5 once more at $k = 2j$. □

3 Proof of Theorem 5

The proof follows the same architecture as the $K = 4$ (first-pair) and $K = 6$ (second-pair) cases treated in the companion papers `2026-04-18-first-q-shifted-pair.tex` and `2026-04-19-second-q-shifted-p` with K now a free even integer parameter. It has three steps: (i) locality of the Hecke element $\Pi_q^{(k)}$; (ii) block-decomposition of D_n with blocks canonically isomorphic to D_K; (iii) multiplicativity of the determinant over the block-diagonal structure.

3.1 Locality

Lemma 7 (Locality of $\Pi_q^{(k)}$)**.** *Let $1 \le k \le K-1$. Then $\Pi_q^{(k)}$ lies in the subalgebra $\langle T_1, T_2, \dots, T_{K-1}\rangle \cong H_q(S_K) \hookrightarrow H_q(S_n)$, for every $n \ge K$. In particular, for any $w \in S_n$, every term $T_{w'}$ appearing in $T_w \cdot \Pi_q^{(k)}$ satisfies $w'(i) = w(i)$ for all $i > K$.*

Proof. $\Pi_q^{(k)} = B_1 B_2 \cdots B_k$ with each $B_j = (1+T_j)\cdots(1+T_1)$ using only generators T_i with $i \le j \le k \le K-1$. Hence $\Pi_q^{(k)} \in \langle T_1, \dots, T_{K-1}\rangle$.

For the second claim, right multiplication by any T_i with $i \le K-1$ maps T_w to a $\mathbb{Z}[q]$-linear combination of T_w and T_{ws_i}; since ws_i swaps the positions i and $i+1$ (both $\le K$), the entries $w(j)$ for $j > K$ are unchanged. Iterating through the factors of $\Pi_q^{(k)}$ preserves this property. □

3.2 Block decomposition of D_n

Lemma 8 (τ_{K-1}-block structure)**.** *Assume $K \ge 4$ is even and $n \ge K$. For each choice of suffix*

$$\sigma' \;=\; (a_{K+1}, a_{K+2}, \dots, a_n) \in \{1, \dots, n\}^{n-K},$$

consisting of $n-K$ distinct integers satisfying $a_{2i-1} > a_{2i}$ for every valid i with both indices $> K$ (the suffix descent-pair constraints), the block

$$V_{\sigma'} \;:=\; \{w \in D_n : \tau_{K-1}(w) = \sigma'\}$$

consists of all descent-paired arrangements of the complementary prefix value-set

$$P_{\sigma'} \;:=\; \{1, 2, \dots, n\} \setminus \{a_{K+1}, \dots, a_n\}$$

into positions $1, 2, \dots, K$. In particular $|V_{\sigma'}| = K!/2^{K/2} = |D_K|$, and the total number of valid suffixes σ' is $|D_n|/|D_K|$, an integer.

Proof. Since K is even, the descent-pair constraints defining D_n split cleanly as

$$\begin{aligned}&\text{prefix: } w(1) > w(2),\ w(3) > w(4),\ \dots,\ w(K-1) > w(K),\\ &\text{suffix: } w(K+1) > w(K+2),\ w(K+3) > w(K+4),\ \dots\end{aligned}$$

(If n is odd, the final position $w(n)$ is unconstrained.) Fixing $\tau_{K-1}(w) = \sigma'$ determines the suffix values and imposes the suffix descent-pair constraints precisely as stated. The prefix values are forced to be the complementary set $P_{\sigma'}$ (of size exactly K), and any descent-paired arrangement of these K values into positions $1, \dots, K$ satisfies the prefix constraints. The number of such arrangements is $K!/2^{K/2} = |D_K|$ (each of the $K/2$ descent pairs halves the count).

Summing $|V_{\sigma'}| = |D_K|$ over all valid suffixes yields $|D_n|$, hence the number of valid suffixes is $|D_n|/|D_K|$; since it is a count of a finite set, it is a non-negative integer. □

Lemma 9 (Matrix-entry equality under block relabelling)**.** *Let $K \ge 4$ be even, $n \ge K$, and σ' a valid suffix. Let $P_{\sigma'} = \{p_1 < p_2 < \cdots < p_K\}$ be the prefix value-set, and let $\phi_{\sigma'} : P_{\sigma'} \to \{1, \dots, K\}$ be the order-preserving bijection sending $p_i \mapsto i$. For any permutation $w \in S_n$ with $\tau_{K-1}(w) = \sigma'$, define the associated permutation*

$$\widehat{w} \;:=\; \phi_{\sigma'} \circ \big(w|_{\{1,\dots,K\}}\big) \;\in\; S_K.$$

The map $w \mapsto \widehat{w}$ is a bijection between $\widetilde{V}_{\sigma'} := \{w \in S_n : \tau_{K-1}(w) = \sigma'\}$ and S_K, restricting to a bijection $V_{\sigma'} \leftrightarrow D_K$.

Then for every $k \in \{1, \dots, K-1\}$ and every $u, v \in V_{\sigma'}$,

$$[T_u](T_v \cdot \Pi_q^{(k)})_{H_q(S_n)} \;=\; [T_{\widehat{u}}](T_{\widehat{v}} \cdot \Pi_q^{(k)})_{H_q(S_K)} \qquad \text{in } \mathbb{Z}[q]. \tag{2}$$

Equivalently, $M^{(k)}[u, v]_{D_n} = M^{(k)}[\widehat{u}, \widehat{v}]_{D_K}$.

Proof. The restriction of $w \mapsto \widehat{w}$ to $V_{\sigma'}$ is a bijection to D_K because $\phi_{\sigma'}$ is order-preserving: descent-pair constraints are defined in terms of the linear order on values, so $w(2i-1) > w(2i) \iff \widehat{w}(2i-1) > \widehat{w}(2i)$ for $i \le K/2$.

We prove the following stronger claim by induction on the length of $\Pi_q^{(k)}$ as a word in the $(1+T_i)$ generators (here $i \le k \le K-1$):

Claim. For every $v \in \widetilde{V}_{\sigma'}$ and every $k \le K-1$, if we write

$$T_v \cdot \Pi_q^{(k)} \;=\; \sum_{w \in \widetilde{V}_{\sigma'}} c_w(q)\, T_w \quad \text{in } H_q(S_n),$$

then with the *same* coefficients $c_w(q) \in \mathbb{Z}[q]$,

$$T_{\widehat{v}} \cdot \Pi_q^{(k)} \;=\; \sum_{w \in \widetilde{V}_{\sigma'}} c_w(q)\, T_{\widehat{w}} \quad \text{in } H_q(S_K).$$

(Recall that $\Pi_q^{(k)} \in \langle T_1, \dots, T_{K-1} \rangle$ by Lemma 7, and that Lemma 7 also implies the support of $T_v \cdot \Pi_q^{(k)}$ lies in $\widetilde{V}_{\sigma'}$.)

Taking $v, u \in V_{\sigma'}$ in the claim and reading off the coefficient of T_u on one side and $T_{\widehat{u}}$ on the other yields (2).

Base case ($k = 0$, trivial product $= 1$). $T_v = T_v$ and $T_{\widehat{v}} = T_{\widehat{v}}$, so the claim holds with $c_v = 1$ and $c_w = 0$ for $w \ne v$.

Induction step. Suppose the claim holds after applying the first ℓ generators $(1+T_{i_1}) \cdots (1+T_{i_\ell})$. We apply the next generator $(1+T_i)$ with $i \le K-1$ on both sides, term by term.

Fix a term T_w appearing on the $H_q(S_n)$-side (so $w \in \widetilde{V}_{\sigma'}$). Since $i \le K-1$ swaps positions i and $i+1$ (both $\le K$), τ_{K-1} is preserved by $w \mapsto ws_i$, so $ws_i \in \widetilde{V}_{\sigma'}$. Moreover $\widehat{ws_i} = (\widehat{w})s_i$, because position-level right-multiplication by s_i commutes with the value-level order-preserving relabelling $\phi_{\sigma'}$.

The Hecke right-action branches on whether $w(i) < w(i+1)$; by order preservation of $\phi_{\sigma'}$, this is equivalent to $\widehat{w}(i) < \widehat{w}(i+1)$. In either case, $T_w \cdot (1+T_i) = \alpha\, T_w + \beta\, T_{ws_i}$ and $T_{\widehat{w}} \cdot (1+T_i) = \alpha\, T_{\widehat{w}} + \beta\, T_{(\widehat{w})s_i}$ with the *same* $(\alpha, \beta) \in \{(1,1), (q,q)\}$ (using $T_w \cdot (1+T_i) = T_w + T_w T_i$ and the standard Hecke rule $T_w \cdot T_i = T_{ws_i}$ or $qT_{ws_i} + (q-1)T_w$). Hence the coefficient expansion on the $H_q(S_n)$-side and the $H_q(S_K)$-side change by the same $\mathbb{Z}[q]$-linear operation, preserving the shared-coefficient structure and completing the induction step. □

3.3 Proof of the theorem

Proof of Theorem 5. **Block-diagonal structure.** We show that $M^{(k)}[D_n, D_n]$ is block-diagonal with respect to the τ_{K-1}-block partition $\{V_{\sigma'}\}_{\sigma' \text{ valid}}$. Concretely, for $v \in V_{\sigma'}$ and $u \in V_{\sigma''}$ with $\sigma' \ne \sigma''$,

$$M^{(k)}[u, v] \;=\; [T_u](T_v \cdot \Pi_q^{(k)}) \;=\; 0,$$

because Lemma 7 guarantees every term $T_{w'}$ in $T_v \cdot \Pi_q^{(k)}$ satisfies $\tau_{K-1}(w') = \tau_{K-1}(v) = \sigma' \ne \sigma'' = \tau_{K-1}(u)$, so the coefficient of T_u vanishes. Hence

$$\det M^{(k)}[D_n, D_n] \;=\; \prod_{\sigma' \text{ valid}} \det M^{(k)}\big|_{V_{\sigma'}}$$

where $M^{(k)}|_{V_{\sigma'}}$ denotes the $V_{\sigma'} \times V_{\sigma'}$ principal submatrix.

Identification of each block with $M^{(k)}[D_K, D_K]$. By Lemma 9, the entry-wise equality (2) holds for every $u, v \in V_{\sigma'}$: under the bijection $V_{\sigma'} \leftrightarrow D_K$, $w \mapsto \widehat{w}$, the (u, v)-entry of $M^{(k)}|_{V_{\sigma'}}$ equals the $(\widehat{u}, \widehat{v})$-entry of $M^{(k)}[D_K, D_K]$. A determinant is invariant under a simultaneous row-and-column relabelling, so $\det M^{(k)}|_{V_{\sigma'}} = \det M^{(k)}[D_K, D_K]$.

Conclusion. Combining the block decomposition and Lemma 8 (which counts the number of valid suffixes as $|D_n|/|D_K|$):

$$\det M^{(k)}[D_n, D_n] \;=\; \prod_{\sigma' \text{ valid}} \det M^{(k)}[D_K, D_K] \;=\; \big(\det M^{(k)}[D_K, D_K]\big)^{|D_n|/|D_K|}.$$

□

4 Application to previously proved pairs

Corollary 10 (Summary of known cases)**.** *Combined with established base cases, Corollary 6 yields:*

1. $j = 1$ ***(all*** $n \geq 4$***, unconditional).*** *The base case* $\det M^{(3)}[D_4, D_4] \cdot \det M^{(1)}[D_4, D_4] = q^6(\det M^{(2)}[D_4, D_4])^2$ *is proved by direct* 6×6 *computation; see* `2026-04-18-first-q-shifted-pair.tex`*.*

2. $j = 2$ ***(all*** $n \geq 6$***, unconditional).*** *The base case at* $K = 6$ *is proved rigorously in* `2026-04-19-second-pair-base-case.tex` *via a q-degree bound* $\deg(F - G) \leq 1444$ *and exact evaluation at* 1449 *distinct integer points* $q_0 \in \{2, \ldots, 1450\}$*.*

3. $j = 3$ ***(all*** $n \geq 8$***, conditional).*** *The base case at* $K = 8$ *is supported by* 9 *mod-prime tests at* $q \in \{2, 3, 5\}$ *and three primes* $p \sim 2^{63}$ *(companion note* `2026-04-20-tsystem-j3-evidence.tex`*), all passing. A rigorous closure via the degree-bound + multi-point method of the* $j = 2$ *base case would require order-of-magnitude more computation; see the companion note for discussion.*

Remark 11 (What the uniform theorem actually does)**.** The companion papers for $j = 1$ and $j = 2$ each proved a *specific* locality reduction (with $K = 4$ and $K = 6$). Theorem 5 unifies those results into a single statement at the level of even K. Its content beyond the union of the companion papers is:

- It verifies that the locality architecture works uniformly for every even K, not just for the two values already checked. In particular, it unconditionally proves the $j = 3$ reduction to the $n = 8$ base case, which the companion papers did not.
- It isolates the K-dependence: only the base case at $n = K$ varies with K; the reduction step is K-independent (modulo substituting K for 4 or 6 throughout).

The proof offers no new insight into *why* the base case holds. A structural (e.g. Desnanot–Jacobi or cluster algebra) derivation of the T-system remains an important open question.

Remark 12 (Non-paired k)**.** Theorem 5 applies to *every* $k \in \{1, \ldots, K-1\}$, not only to the three values $\{2j-1, 2j, 2j+1\}$ needed for the T-system. Consequently the factorisation

$$\det M^{(k)}[D_n, D_n] \;=\; \big(\det M^{(k)}[D_K, D_K]\big)^{|D_n|/|D_K|}$$

holds for every $n \geq K$ and every $k \leq K - 1$ with K even. This is already a non-trivial structural statement: a single small-n computation determines the determinant at all larger n via a pure exponent.

5 Integrality of $|D_n|/|D_K|$

We collect the integrality statement used in Lemma 8; it is built into the combinatorial argument but we record the direct computational proof for completeness.

Proposition 13. *For every even $K \geq 2$ and every $n \geq K$,*

$$\frac{|D_n|}{|D_K|} = \frac{n!}{K! \cdot 2^{\lfloor n/2 \rfloor - K/2}}$$

is a positive integer.

Proof. The quantity equals the number of valid suffixes $\sigma' = (w(K+1), \ldots, w(n))$ obtained from elements $w \in D_n$, by Lemma 8. As a cardinality of a finite set it is a non-negative integer. Positivity for $n \geq K$ is immediate (the suffix is empty when $n = K$, giving 1; and for each $n > K$ one can exhibit at least one valid suffix by descent-pairing the largest $n - K$ values in decreasing order within consecutive pairs). □

Remark 14 (Algebraic form)**.** Equivalently, for $n = K + 2t$ (even) or $n = K + 2t + 1$ (odd) with $t \geq 0$,

$$|D_n|/|D_K| = \begin{cases} \dfrac{(K+2t)!}{K!\,2^t}, & n = K + 2t, \\ \dfrac{(K+2t+1)!}{K!\,2^t}, & n = K + 2t + 1. \end{cases}$$

Integrality follows from $n!/K!$ containing at least t even integers (namely $K+2, K+4, \ldots, K+2t$).

6 Numerical sanity checks

The cases $K = 4$ and $K = 6$ of Theorem 5 have been verified in the companion papers for all n up to 7 (first pair) and 7 (second pair) respectively. We record here a confirmation at the first new value of K:

Example 15 ($K = 8$, cross-check at $n = 9, 10$)**.** From the $K = 8$ block decomposition, Theorem 5 predicts

$$\det M^{(k)}[D_9, D_9] = \left(\det M^{(k)}[D_8, D_8]\right)^9 \quad \text{for } k \leq 7,$$
$$\det M^{(k)}[D_{10}, D_{10}] = \left(\det M^{(k)}[D_8, D_8]\right)^{45} \quad \text{for } k \leq 7.$$

(The exponents $9, 45$ are $|D_9|/|D_8| = 22680/2520$ and $|D_{10}|/|D_8| = 113400/2520$ respectively.)

A direct mod-prime check of the $j = 3$ T-system identity at $n = 9$ or $n = 10$ is computationally prohibitive via full matrix assembly ($|D_9| = 22680$, $|D_{10}| = 113400$), but the identity at $n = K = 8$ has been cross-checked at 9 mod-prime tests (companion note), so the theorem transports the check unchanged to larger n.

7 Notes on the structural open question

Theorem 5 is, in essence, a structural statement about the *reduction step* in the T-system: the infinite family of identities for $n \geq K$ collapses to a single identity at $n = K$. It does not address the harder question of why the base-case identity itself holds.

Three natural routes to a uniform proof of the base case (as K varies) are:

1. **Master matrix via Dodgson condensation.** Find a single master matrix M^* on some enlarged index set whose principal submatrices recover $M^{(k)}[D_K]$ for successive k, such that the T-system identity becomes a Desnanot–Jacobi (Plücker) identity with vanishing cross-minors. To date no such master matrix has been identified; all $M^{(k)}$ live on the same-sized index set D_K, ruling out the simplest Dodgson structure.

2. **Cluster algebra structure.** T-systems arise naturally in cluster algebras (Kirillov–Reshetikhin modules, quantum affine $\mathfrak{sl}_2$). A cluster seed whose mutation sequence reproduces the staircase ratios R_k would give the T-system automatically.

3. **Eigenvalue factorisation + Frobenius reciprocity.** The eigenvalue positivity theorem (Apr 17) gives positive palindromic eigenvalues on each irrep, but the determinant $\det M^{(k)}[D_n]$ does not obviously factor over irreps because D_n is not $H_q(S_n)$-stable. A finer decomposition adapted to the paired-descent structure might yield per-irrep T-systems whose product gives the whole.

None of these is closed at present. The uniform locality reduction proved above isolates exactly which problem remains.

Acknowledgements

Companion papers:

- `2026-04-18-first-q-shifted-pair.tex` — $K = 4$ instance.
- `2026-04-19-second-q-shifted-pair.tex` — $K = 6$ instance (locality reduction).
- `2026-04-19-second-pair-base-case.tex` — rigorous closure of the $K = 6$ base case.
- `2026-04-20-tsystem-j3-evidence.tex` — mod-prime evidence for the $K = 8$ base case.